

Motor Control Board

Variant: DRAFT

2026-08-20

Rev + (Unreleased)

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TOP VIEW

kibot_Mimage_png_3d_viewer_angled_top

BOTTOM VIEW

kibot_Mimage_png_3d_viewer_angled_bottom

DESIGN CONSIDERATIONS

DESIGN NOTE:
Example text for
informational design
notes.

DESIGN NOTE:
Example text for
debug notes.

DESIGN NOTE:
Example text for
cautionary design
notes.

DESIGN NOTE:
Example text for
critical design
notes.

LAYOUT NOTE:
Example text for
critical layout
guidelines.

NOTES

Comment

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Block Diagram

File: Block Diagram.kicad_sch

Revision History

File: Revision History.kicad_sch

Project Architecture

File: Project Architecture.kicad_sch

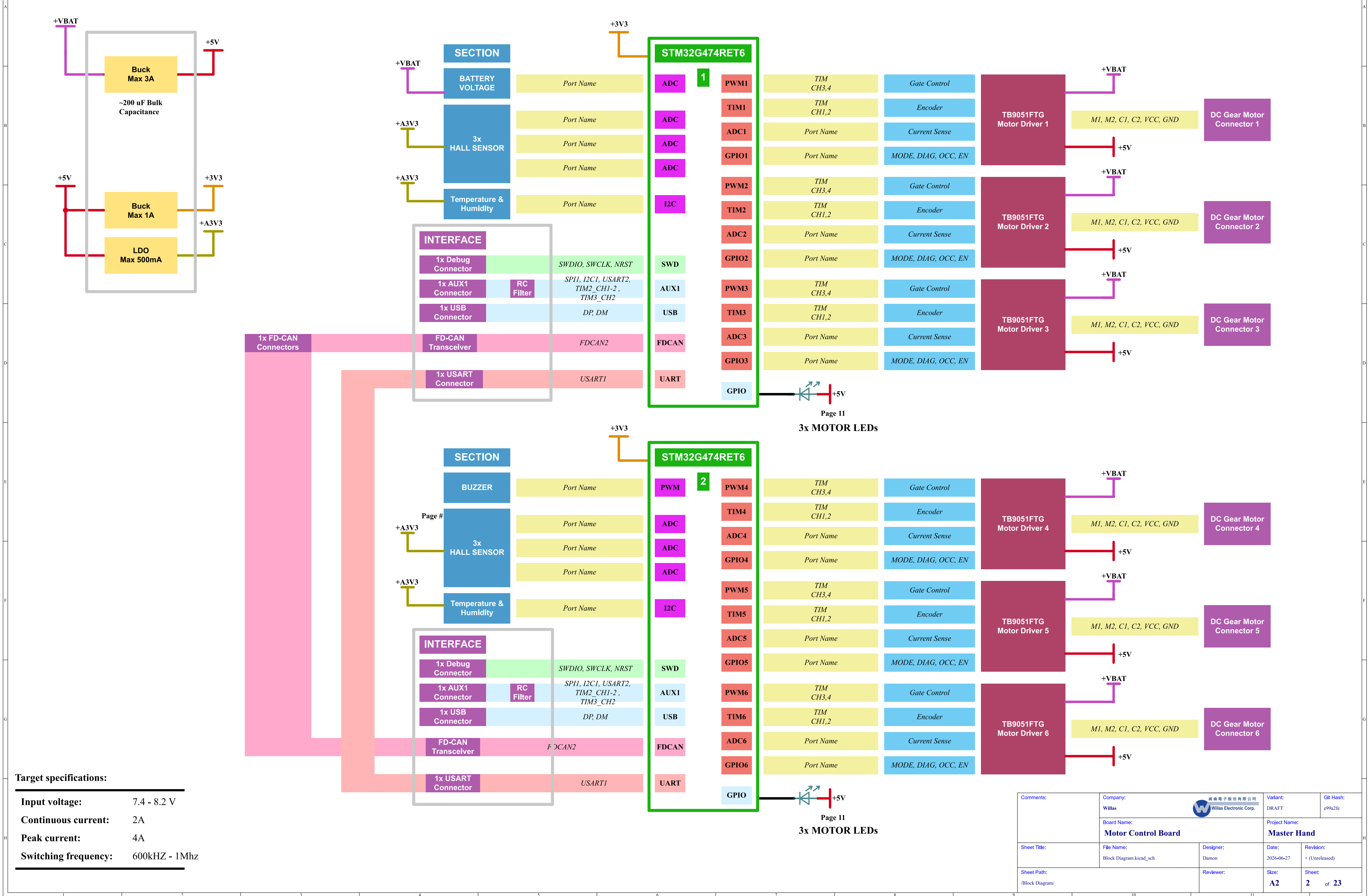
Power - Sequencing

File: Power - Sequencing.kicad_sch

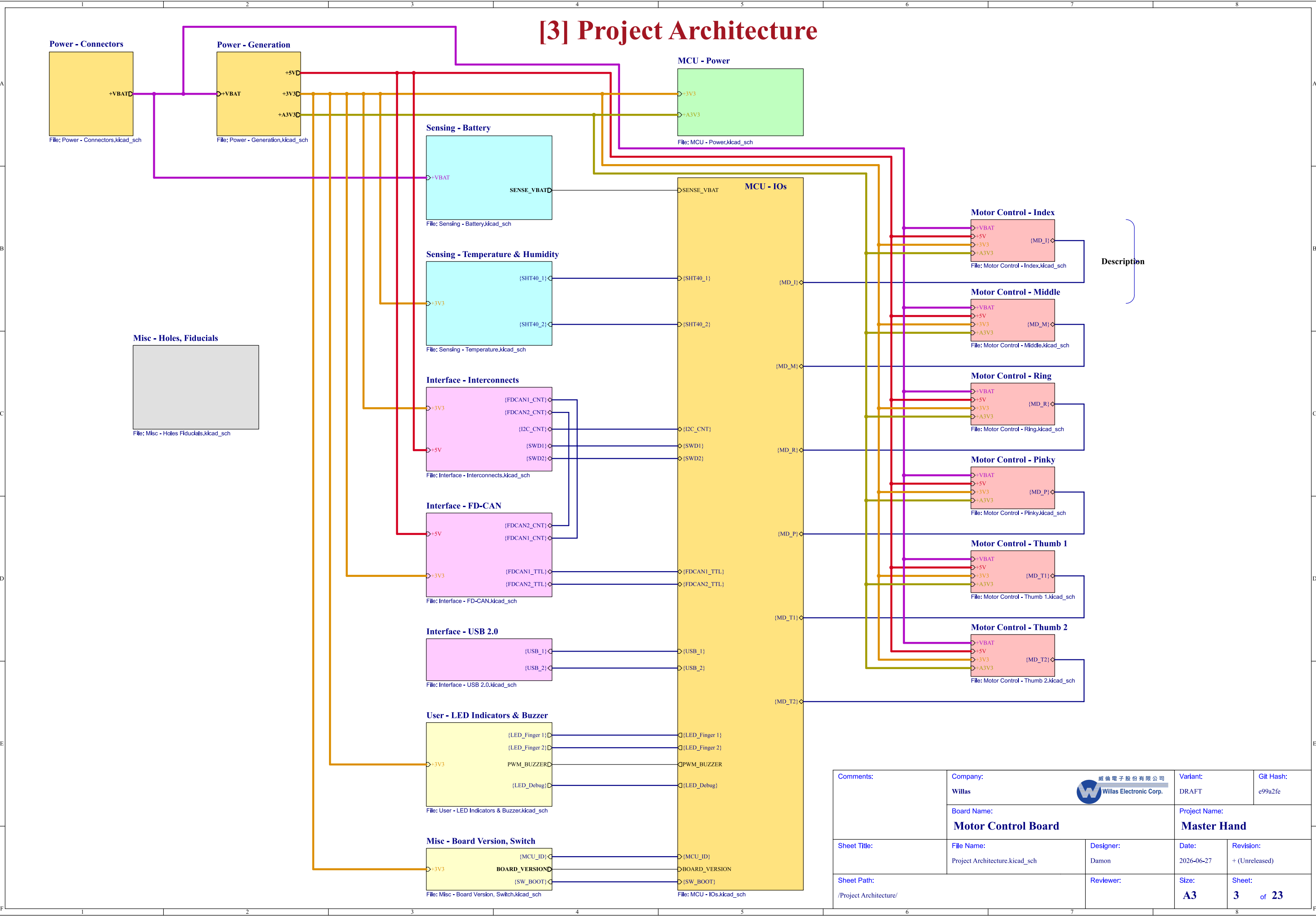
Date: 20-Aug-2026

Comments:	Company: Willas		W Willas Electronic Corp.		Variant: DRAFT	Git Hash: e99a2fe
	Board Name: Motor Control Board				Project Name: Master Hand	
Sheet Title:	File Name: Master_Hand_Kicad.kicad_sch		Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
Sheet Path: /			Reviewer:	Size: A3	Sheet: 1 of 23	

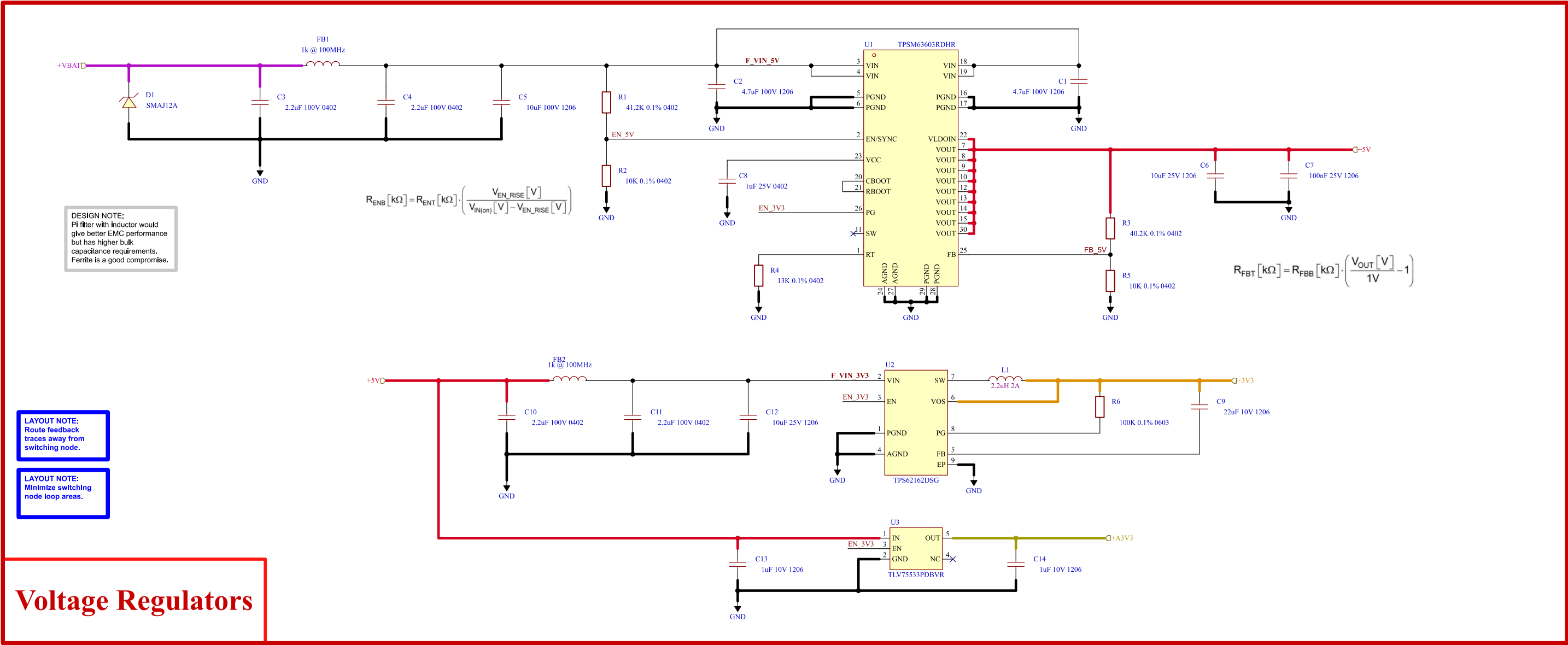
[2] Block Diagram



[3] Project Architecture



[4] Power - Generation



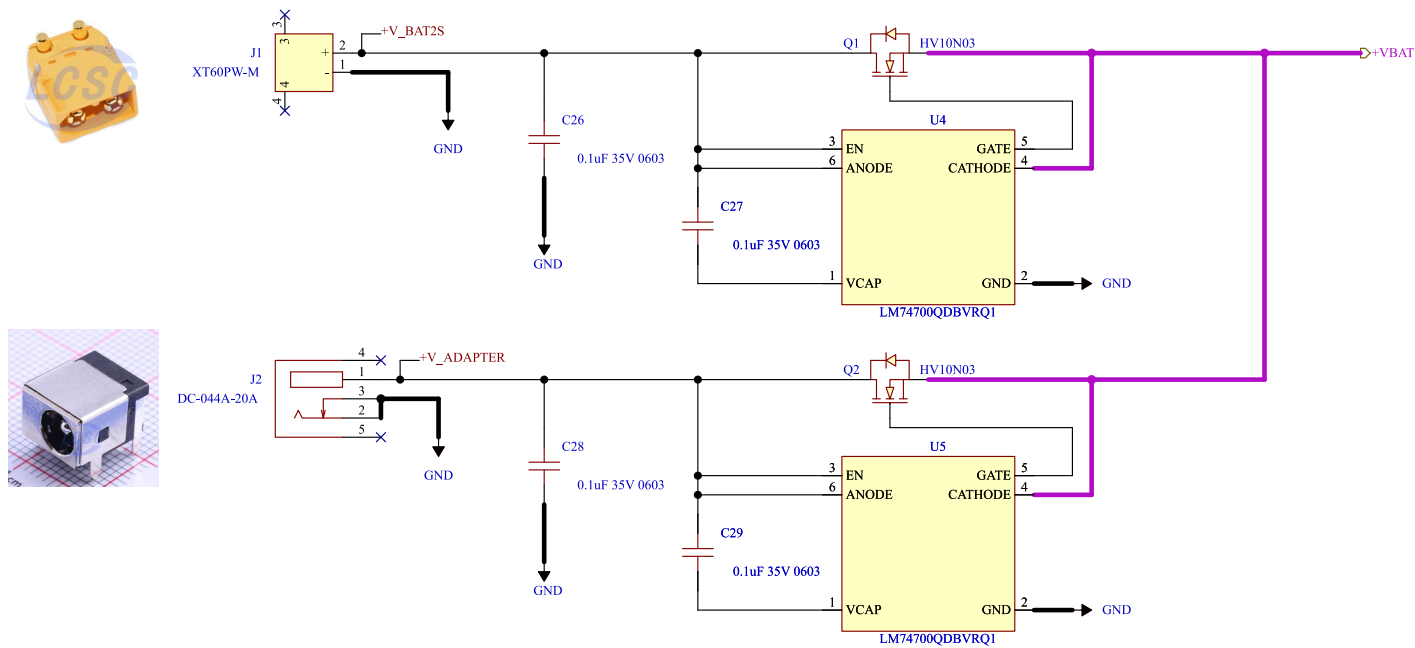
Voltage Regulators

DC BULK

Test Points

Comments:	Company: Willas		 威倫電子股份有限公司 Willas Electronic Corp.		Variant: DRAFT	Git Hash: e99a2fc
	Board Name: Motor Control Board			Project Name: Master Hand		
Sheet Title:	File Name: Power - Generation.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)		
Sheet Path:	/Project Architecture/Power - Generation/		Reviewer:	Size: A3	Sheet: 4 of 23	

[5] Power - Connectors

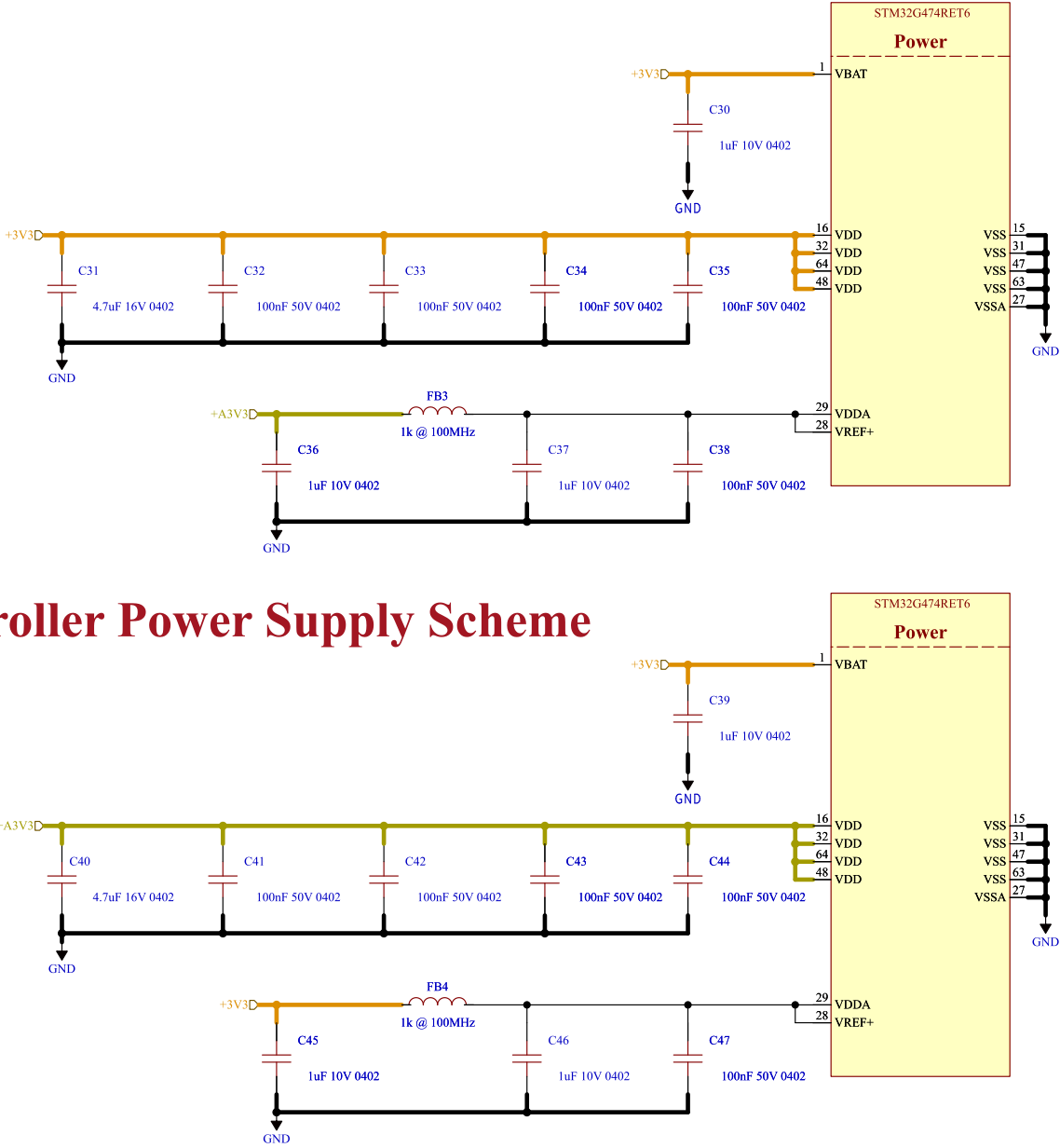


Power Connectors - Path Control

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	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Power - Connectors.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Power - Connectors/		Reviewer:	Size: A4	Sheet: 5 of 23

Microcontroller Power Supply Scheme

LAYOUT NOTE:
Place each capacitor close
to the relevant pin.
Minimize loop inductance.



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	Board Name: Motor Control Board			Project Name: Master Hand		
Sheet Title:	File Name: MCU - Power.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)		
Sheet Path: /Project Architecture/MCU - Power/	Reviewer:		Size: A3	Sheet: 6 of 23		

Battery Voltage Sense

PWM_BUZZER PWM_BUZZER

Buzzer

Diagram illustrating the connection of MD_I to various modules:

- MD_I is connected to PWMA_I, PWMB_I, EN_I, DIAG_I, OCM_I, ECA_I, ECB_I, and ECC_I.

The diagram shows two input signals, `I2C_SDA` and `I2C_SCL`, connected to the `I2C_CNT` pin of the `MCB to MB - I2C` module. The signals are represented by blue lines that enter from the left and connect to the pin on the right.

Diagram illustrating a 5Mbps FD-CAN Bus configuration. The bus is shown as a horizontal line with termination symbols (diamonds) at both ends, labeled {FDCAN1_TTL}. Two nodes are connected to the bus:

- Node 1 (Top): Labeled FDCAN1_RX and FDCAN1_TX.
- Node 2 (Bottom): Labeled FDCAN2_RX and FDCAN2_TX.

GPIO-LED_I
GPIO-LED_T1
GPIO-LED_T2

{LED_Finger 1}

GPIO-LED_P
GPIO-LED_R
GPIO-LED_M

{LED_Finger 2}

GPIO-LED_Debug_1
GPIO-LED_Debug_2

{LED_Debug}

LEDs

A diagram of a switch component. It has two input lines on the left, labeled BOOT_1 and BOOT_2. These lines enter a trapezoidal shape representing the switch. From the right side of the switch, a single output line extends to the right, ending in a yellow circle and labeled {SW_BOOT}.

The diagram illustrates the internal wiring of a USB port. On the left, three horizontal lines represent the USB signals: USB-DP_1, USB-DM_1, and VBUS_DET1. These lines enter a vertical connector block. From the top of this block, a single line extends to the right, ending in a yellow circle with a blue outline, labeled **{USB_1}**. On the right, a similar setup is shown with three horizontal lines: USB-DP_2, USB-DM_2, and VBUS_DET2. These enter another vertical connector block, with a single line extending to the right to a yellow circle with a blue outline, labeled **{USB_2}**.

USB PORT

nts:	Company: Willas		Willas Electronic Corp.		Variant: DRAFT	Git Hash: e99a2fc
	Board Name: Motor Control Board				Project Name: Master Hand	
File:	File Name: MCU - IOs.kicad_sch		Designer: Damon		Date: 2026-06-27	Revision: + (Unreleased)
Path: Architecture/MCU - IOs/			Reviewer:		Size: A3	Sheet: 7 of 23

Comments:

Company:	
----------	--

Willas

Board Name:

Motor Control Board

File Name:	
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MCU - IOs.kicad_sch

Variant:

DRAFT

Project Name:

Master Hand

Date:

2026-06-27

Size:

A3 7 of 23

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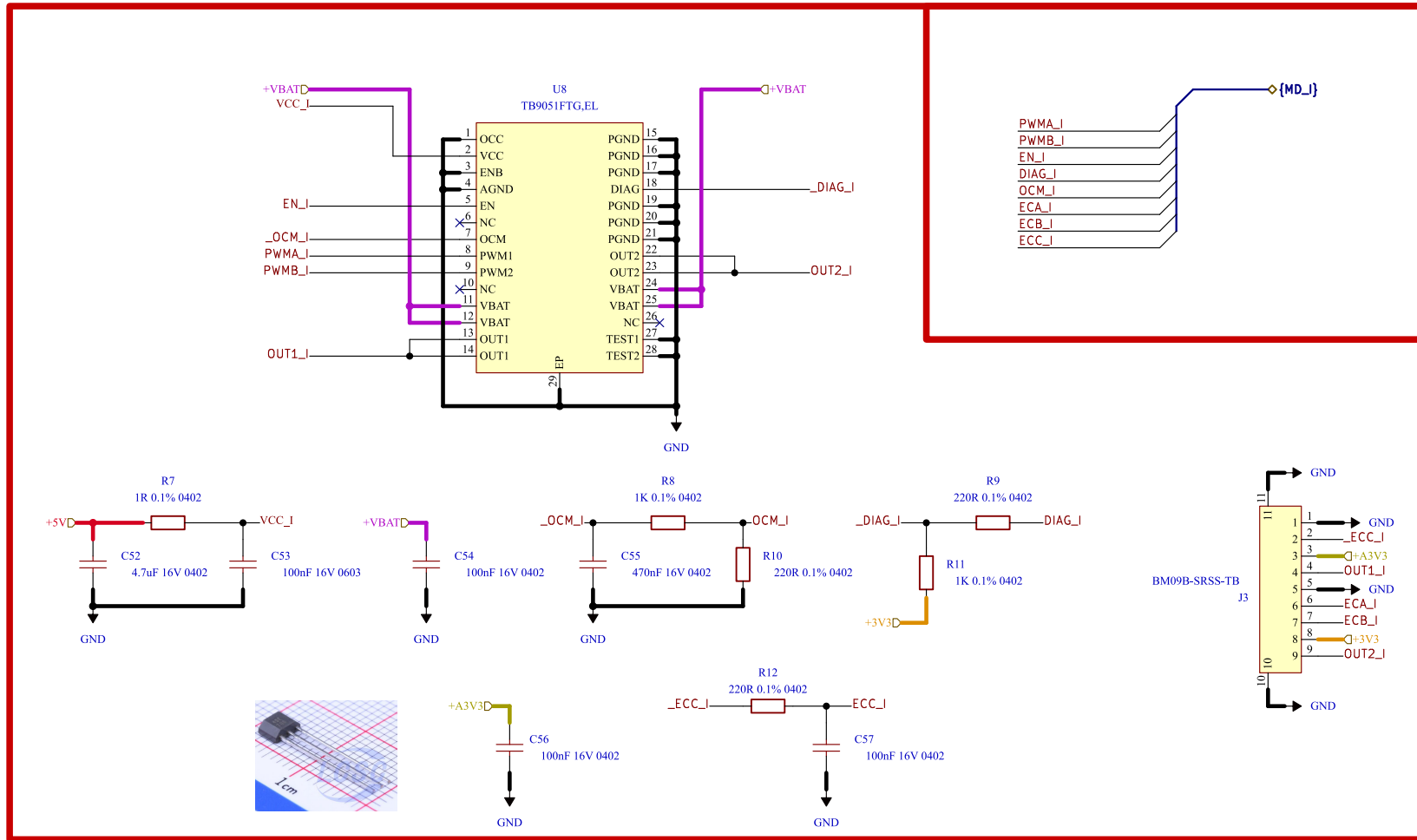
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23

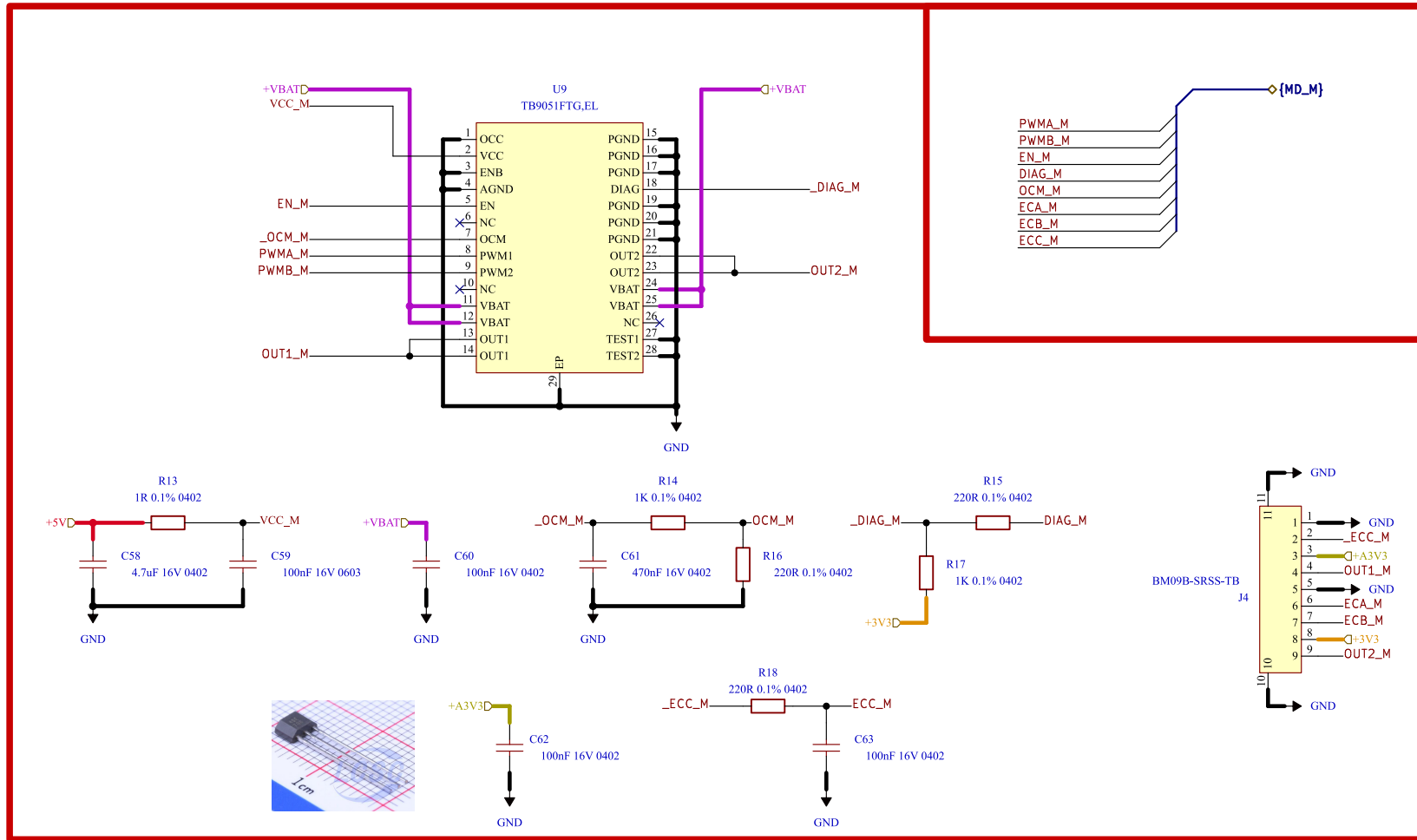
LAYOUT NOTE:
Footprint deviates from
datasheet and IPC7351 to
allow for soldermask sliver
between pads according to
PCBWay design rules.

[8] Motor Control - Index



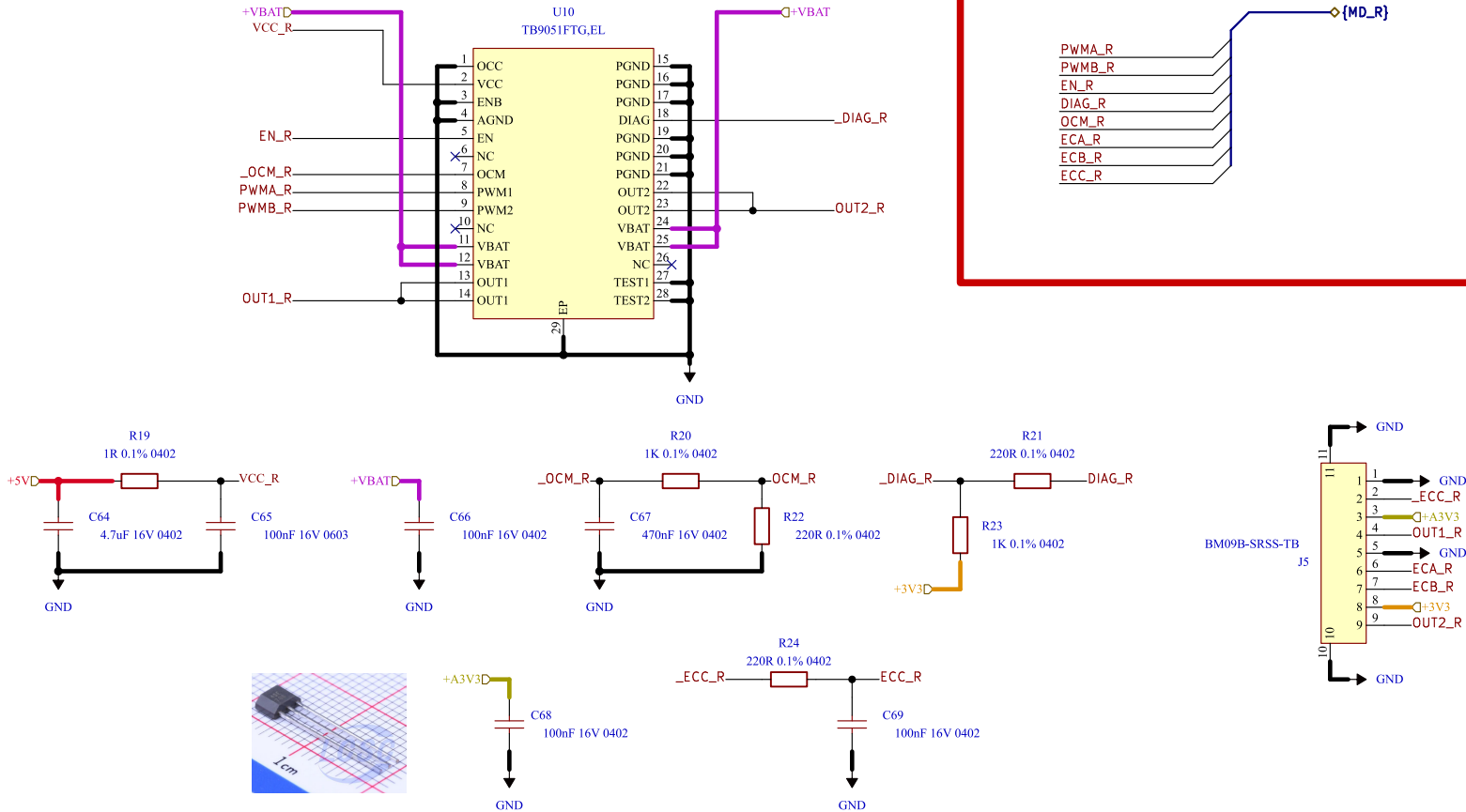
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	Board Name: Motor Control Board			Project Name: Master Hand	
Sheet Title:	File Name: Motor Control - Index.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
Sheet Path: /Project Architecture/Motor Control - Index/		Reviewer:	Size: A4	Sheet: 8 of 23	

[9] Motor Control - Middle



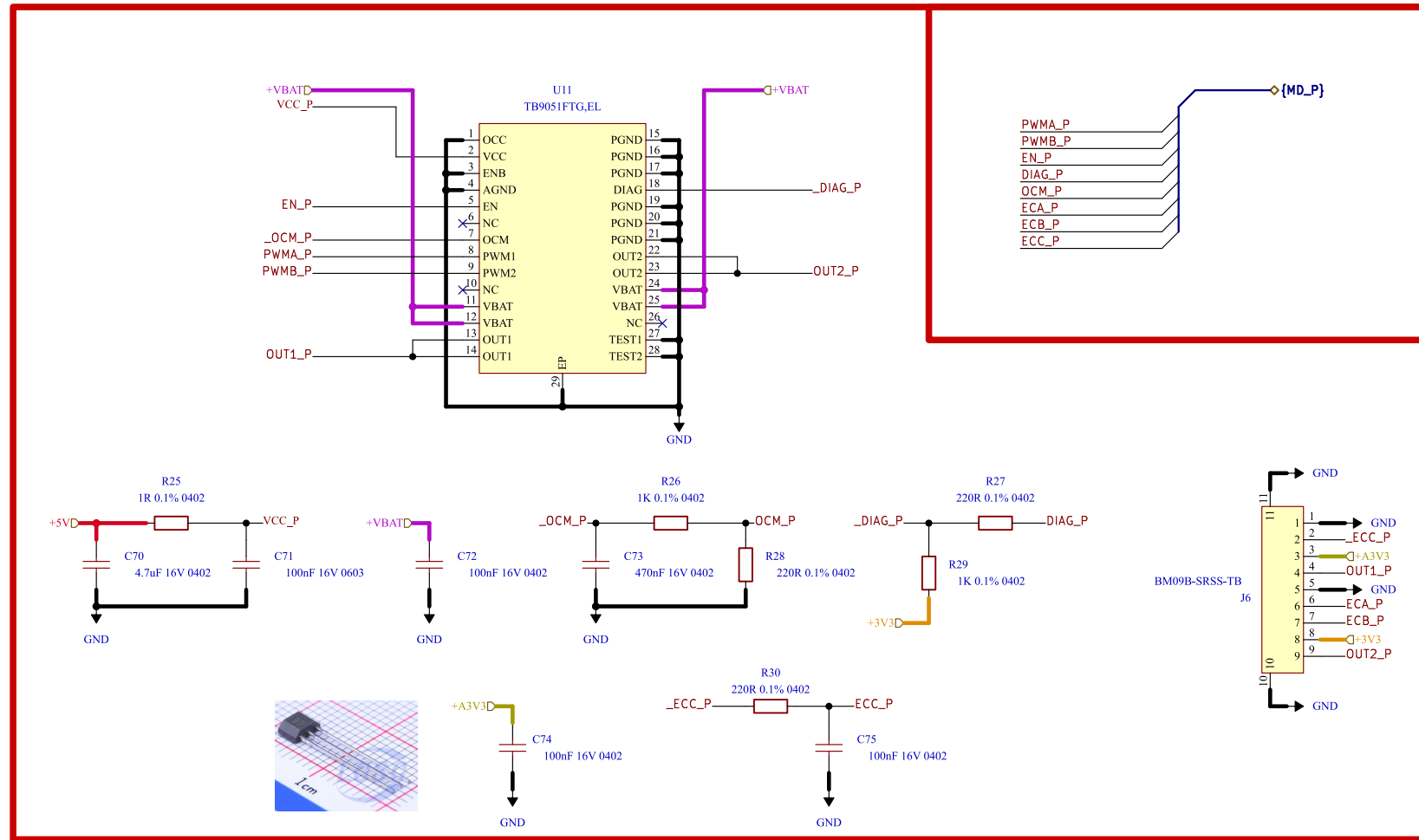
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	Board Name: Motor Control Board		Project Name: Master Hand		
Sheet Title:	File Name: Motor Control - Middle.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
Sheet Path: /Project Architecture/Motor Control - Middle/		Reviewer:	Size: A4	Sheet: 9 of 23	

[10] Motor Control - Ring



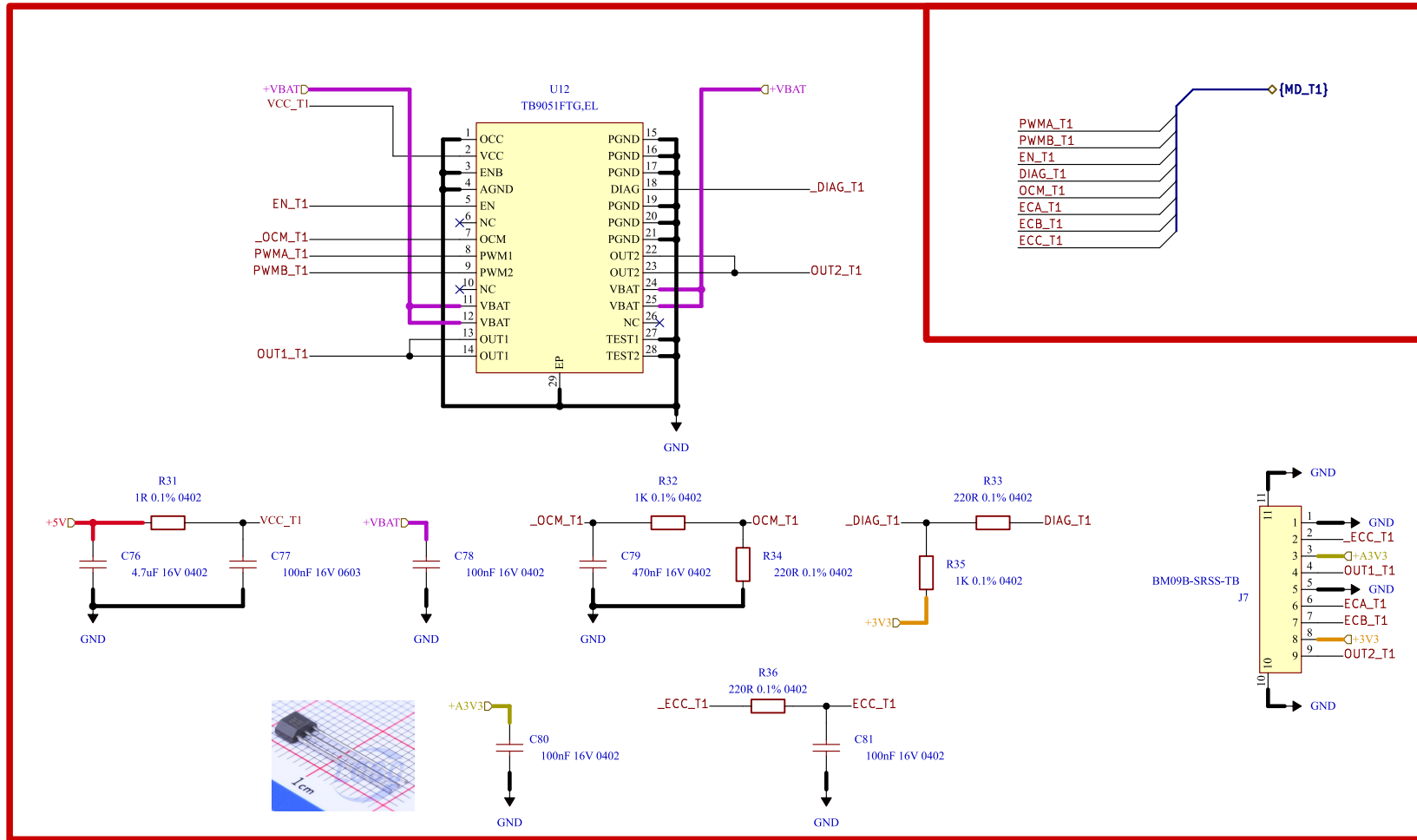
Comments:	Company: Willas		威倫電子股份有限公司 Willas Electronic Corp.		Variant: DRAFT	Git Hash: e99a2fc
	Board Name: Motor Control Board			Project Name: Master Hand		
Sheet Title:	File Name: Motor Control - Ring.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)		
Sheet Path: /Project Architecture/Motor Control - Ring/		Reviewer:	Size: A4	Sheet: 10 of 23		

[11] Motor Control - Pinky



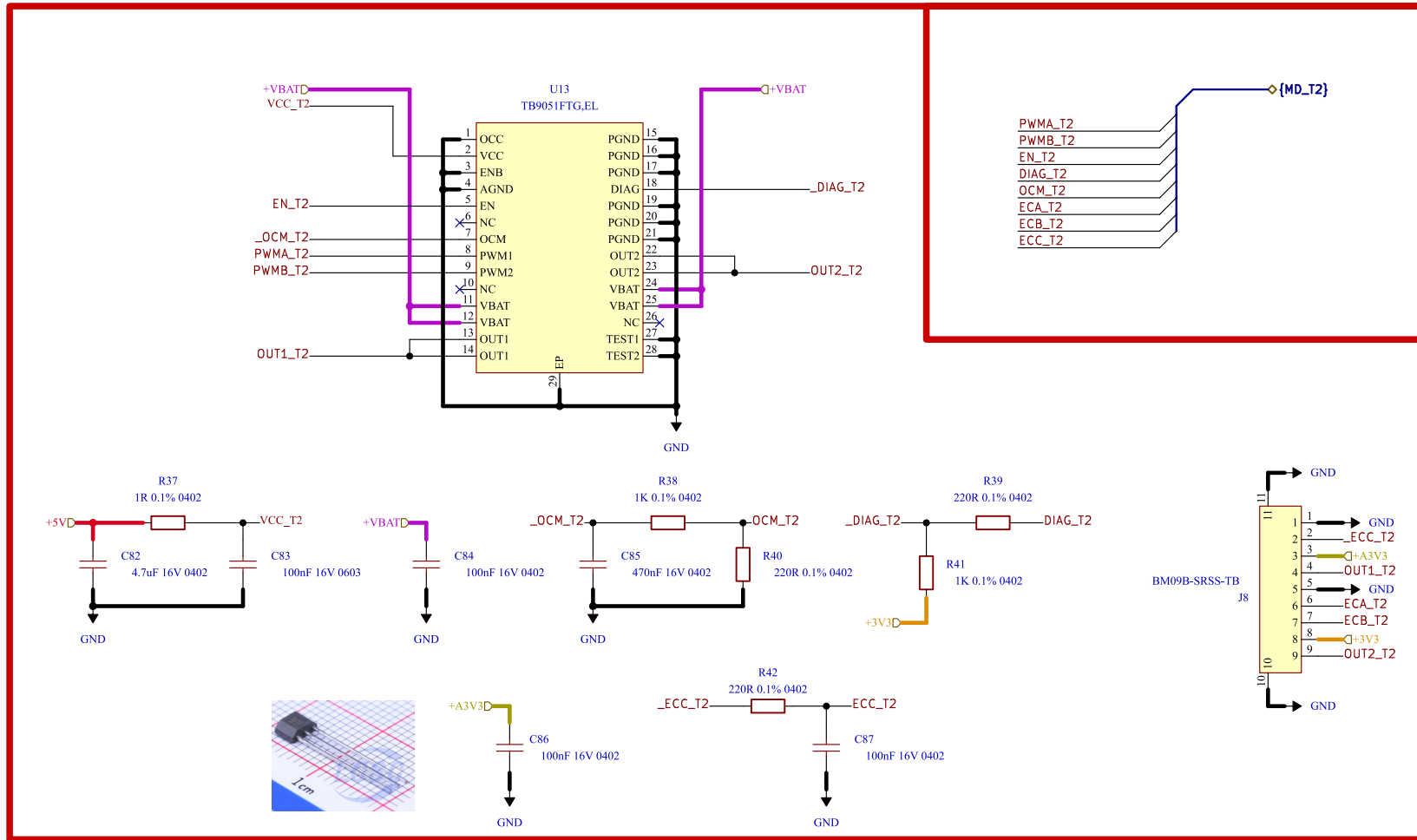
Comments:	Company: Willas  威倫電子股份有限公司 Willas Electronic Corp.		Variant: DRAFT	Git Hash: e99a2fe
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Motor Control - Pinky.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Motor Control - Pinky/		Reviewer:	Size: A4	Sheet: 11 of 23

[12] Motor Control - Thumb 1



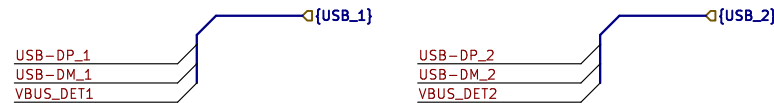
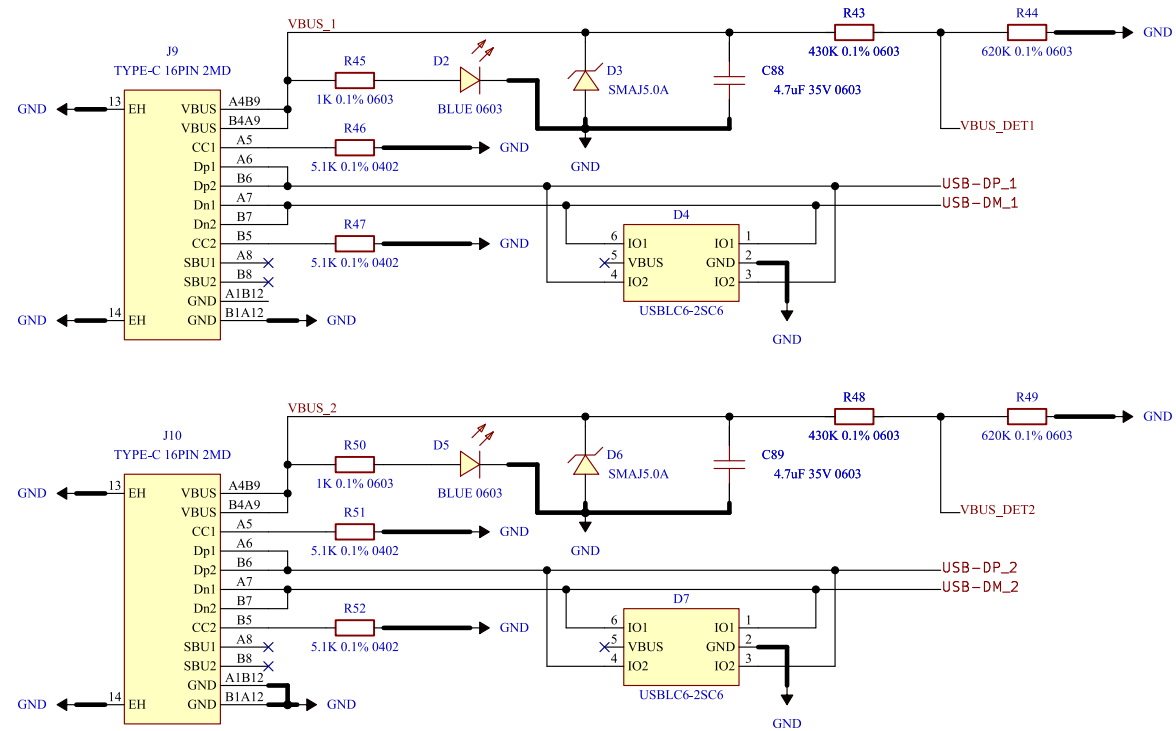
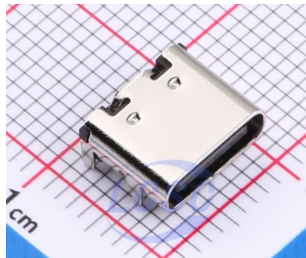
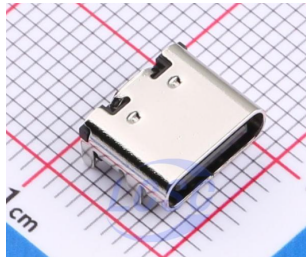
Comments:	Company: Willas Willas Electronic Corp.		Variant: DRAFT	Git Hash: e99a2fe
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Motor Control - Thumb 1.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Motor Control - Thumb 1/		Reviewer:	Size: A4	Sheet: 12 of 23

[13] Motor Control - Thumb 2



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	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Motor Control - Thumb 2.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Motor Control - Thumb 2/		Reviewer:	Size: A4	Sheet: 13 of 23

[14] Interface - USB 2.0



DESIGN NOTE:

- [1] VBUS NOT connected to +5V/+3V3. Board is self-powered(LiPo). Sense + LED only. No backfeed to host.
- [2] LED direct from VBUS @ 1mA. USB2.0 §7.2.3 suspend limit 2.5mA. LED 1mA + divider 33uA = 1.03mA (41%). Do NOT raise to 5mA.
- [3] VBUS_DET required - G474 has no dedicated VBUS pin. Self-powered device must not pull up D+ when VBUS=0. FW controls DPPU bit in USB_BCDR. 68k/100k: VBUS 4.40V -> 2.60V > VIH 2.31V. Debounce 100ms.
- [4] D+/D- direct to PA12/PA11. No 22R series. No external 1k5 pull-up (internal RPUI 900-1500R).
- [5] USB clock: PLL-Q 48MHz from HSE 8MHz. ST DFU bootloader uses HS148+CRS - flashable without crystal.
- [6] Both USB ports fully isolated - no shared nets.
- [7] GROUND LOOP WARNING: USB from PC ties host GND to power GND (4.6A peak @20kHz). Use isolator (ADuM3160) or isolated USB-UART when debugging with motors running.

Comments:

Company:

Willas



Variant:

DRAFT

Git Hash:

c99a2fe

Board Name:

Motor Control Board

Project Name:

Master Hand

Sheet Title:

File Name:

Interface - USB 2.0.kicad_sch

Designer:

Damon

Date:

2026-06-27

Revision:

+(Unreleased)

Sheet Path:

/Project Architecture/Interface - USB 2.0/

Reviewer:

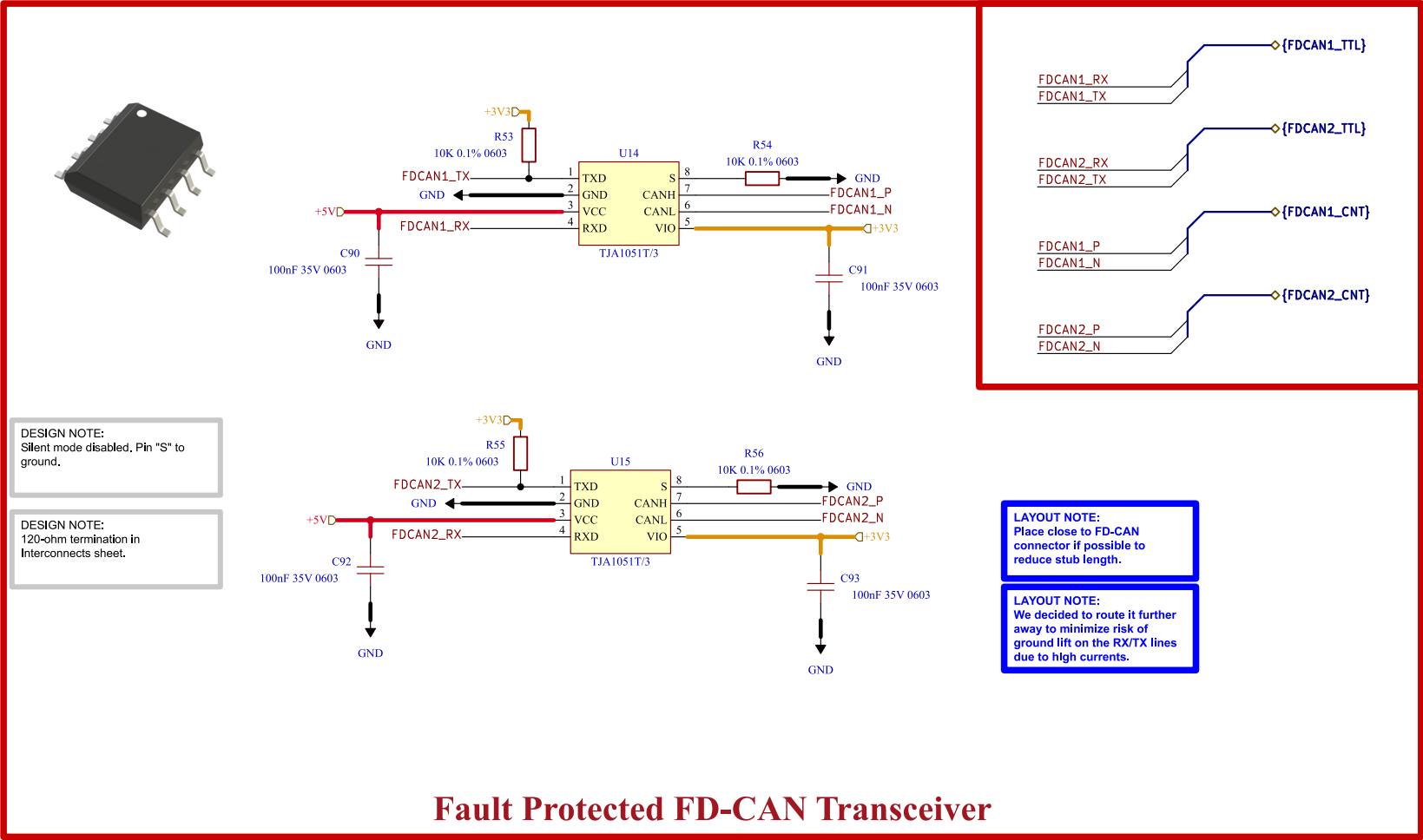
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A4

Sheet:

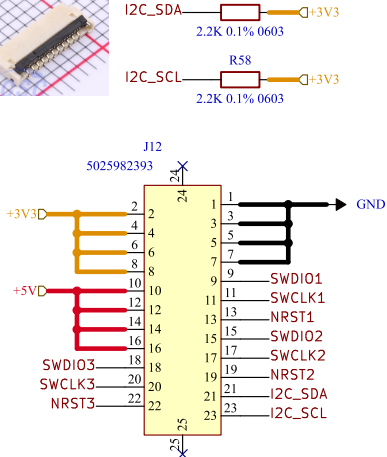
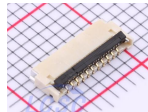
14 of 23

[15] Interface - FD-CAN

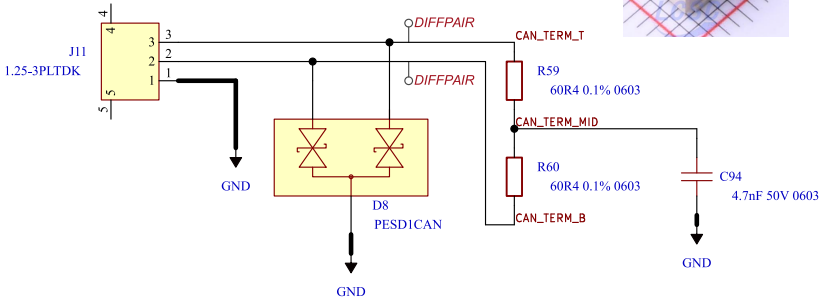


Comments:	Company: 威倫電子股份有限公司 Willas		Variant: DRAFT	Git Hash: e99a2fe
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Interface - FD-CAN.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Interface - FD-CAN/		Reviewer:	Size: A4	Sheet: 15 of 23

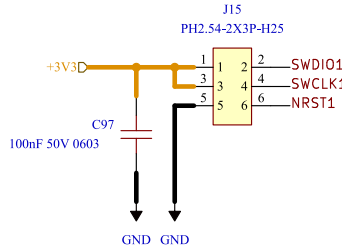
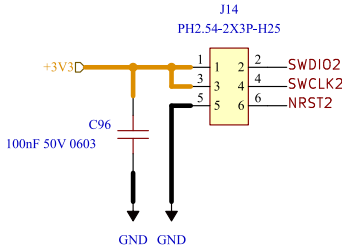
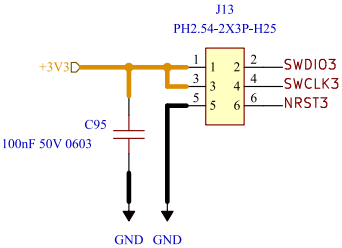
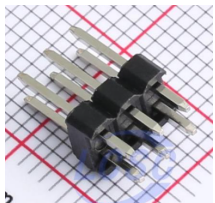
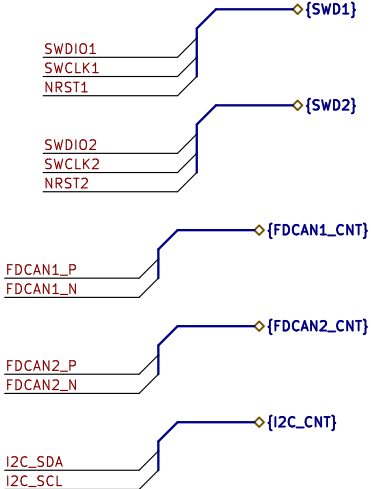
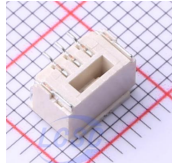
[16] Interface - Interconnects



FFC Connector MCB & MB



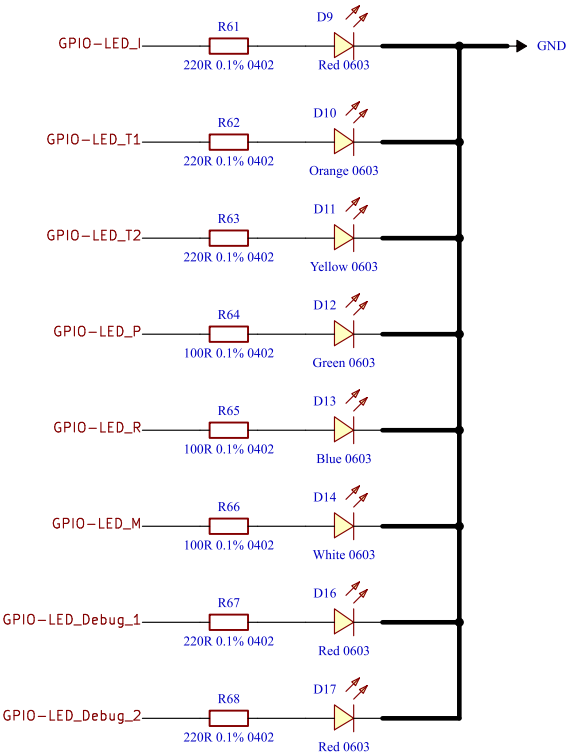
FD-CAN Connector and Termination



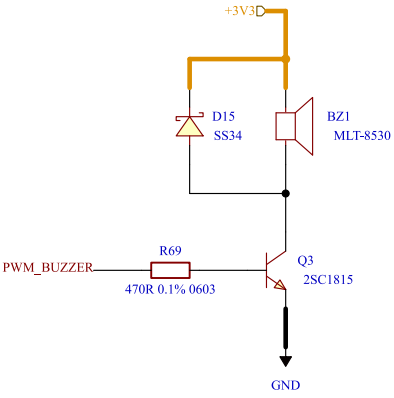
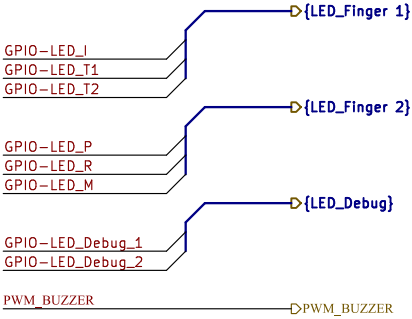
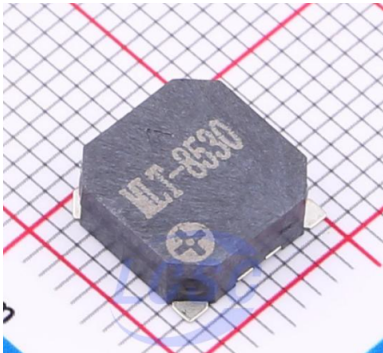
SWD Debug Connector

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	Board Name: Motor Control Board				Project Name: Master Hand	
Sheet Title:	File Name: Interface - Interconnects.kicad_sch		Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
Sheet Path: /Project Architecture/Interface - Interconnects/			Reviewer:	Size: A4	Sheet: 16 of 23	

[17] User - LED Indicators & Buzzer

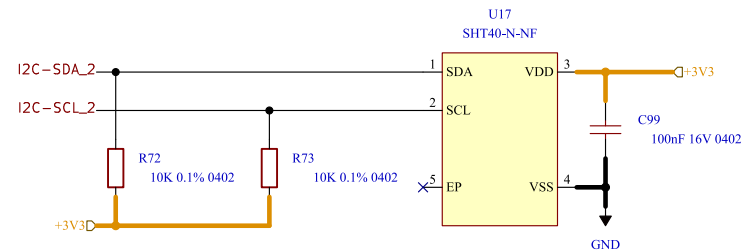
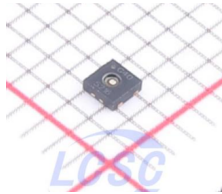
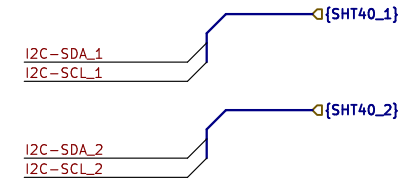
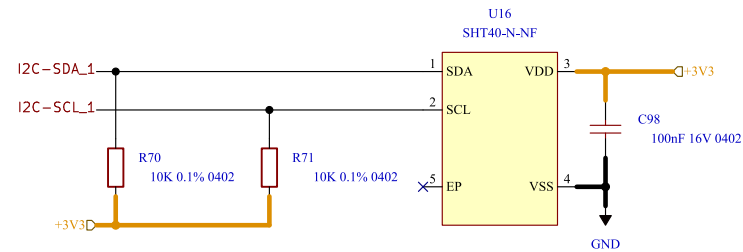
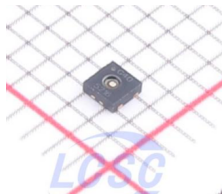


LED Color	Typical Vf Range
Red	1.8 to 2.1
Amber	2 to 2.2
Orange	1.9 to 2.2
Yellow	1.9 to 2.2
Green	2 to 3.1
Blue	3 to 3.7
White	3 to 3.4



Comments:	Company: Willas	Variant: DRAFT	Git Hash: e99a2fe	
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: User - LED Indicators & Buzzer.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/User - LED Indicators & Buzzer/	Reviewer:	Size: A4	Sheet: 17 of 23	

[18] Sensing - Temperature & Humidity



Comments:	Company: 威倫電子股份有限公司 Willas Electronic Corp.		Variant: DRAFT		Git Hash: e99a2fe
	Board Name: Motor Control Board		Project Name: Master Hand		
Sheet Title:	File Name: Sensing - Temperature.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
Sheet Path: /Project Architecture/Sensing - Temperature & Humidity/		Reviewer:	Size: A4	Sheet: 18 of 23	

[19] Sensing - Battery

DESIGN NOTE:
Battery voltage sensing divider for 2S LiPo battery.

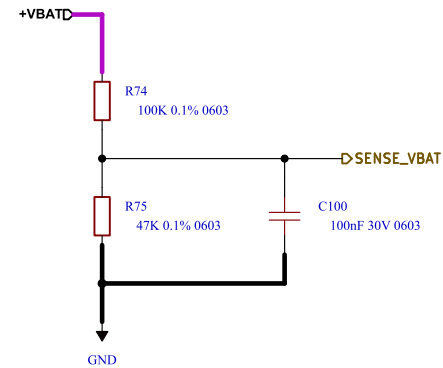
R1 = 100kΩ (Top)
R2 = 47kΩ (Bottom)

Divider Ratio:
 $V_{ADC} = V_{BAT} \times (47 / (100 + 47))$
 $V_{ADC} = V_{BAT} \times 0.3197$

2S LiPo:
Max battery voltage (8.4V): 2.69 V
Nominal voltage (7.4V): 2.37 V
Low battery voltage (6.4V): 2.05 V
Cutoff voltage (6.0V): 1.92 V

Maximum divider current:
 $I = 8.4V / 147k\Omega = 57 \mu A$

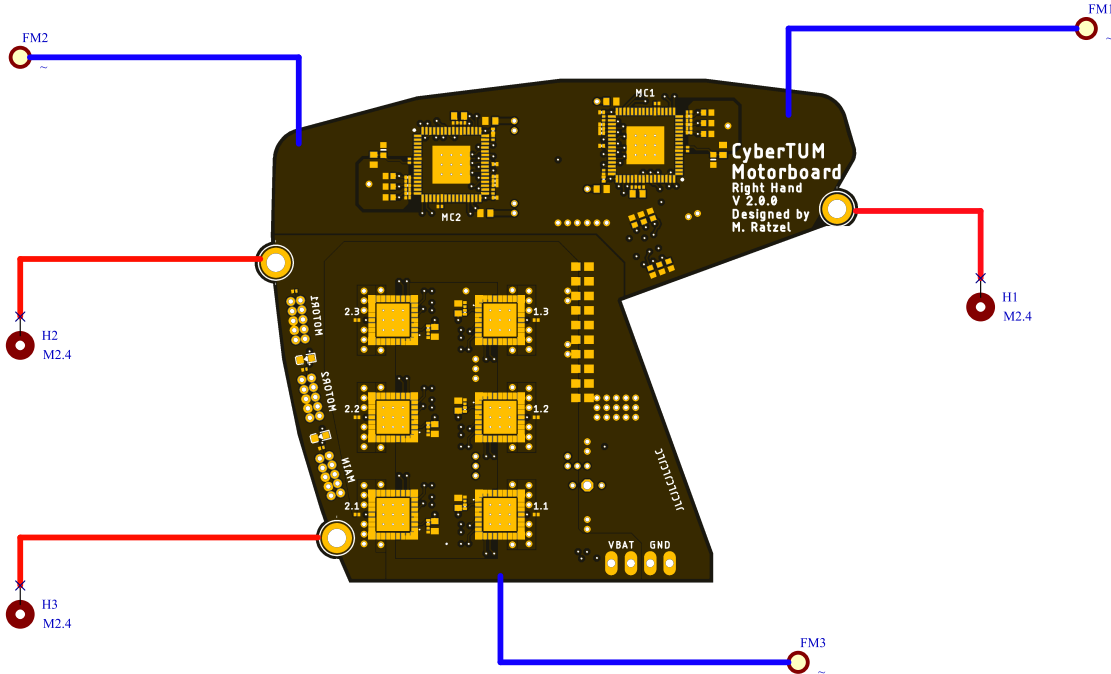
Notes:
- Designed for 3.3V ADC input.
- Add 100nF capacitor at ADC input for noise filtering.
- Use 1% resistors or better.
- Use low temperature coefficient resistors ($\leq 100 \text{ ppm}/^\circ\text{C}$) for improved measurement accuracy.



Battery Voltage Sensing

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	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Sensing - Battery.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Project Architecture/Sensing - Battery/		Reviewer:	Size: A4	Sheet: 19 of 23

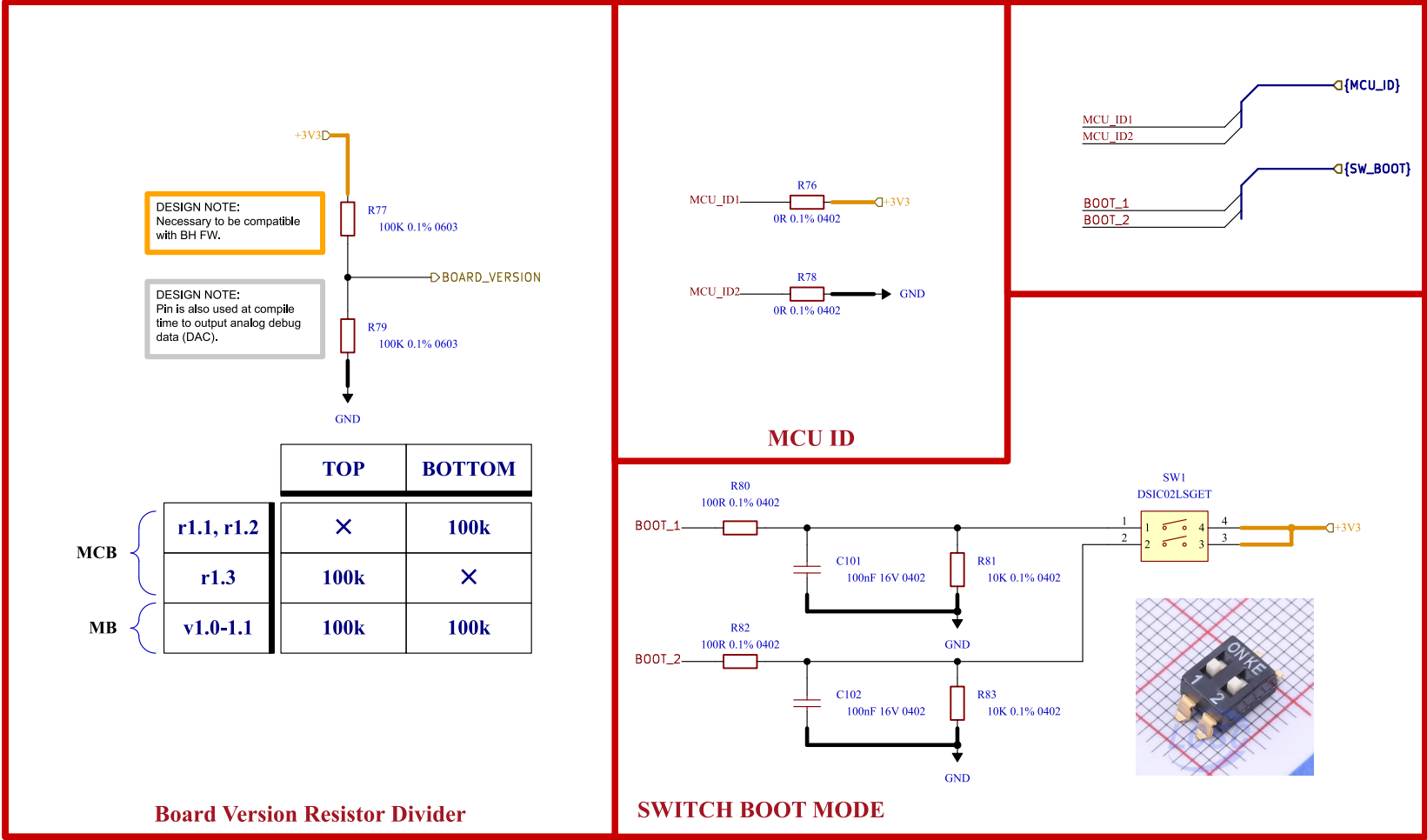
[20] Misc - Holes, Fiducials



Mounting Holes and Fiducial Markers

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Sheet Title:	File Name: Misc - Holes Fiducials.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)		
Sheet Path: /Project Architecture/Misc - Holes, Fiducials/			Reviewer:	Size: A4	Sheet: 20 of 23	

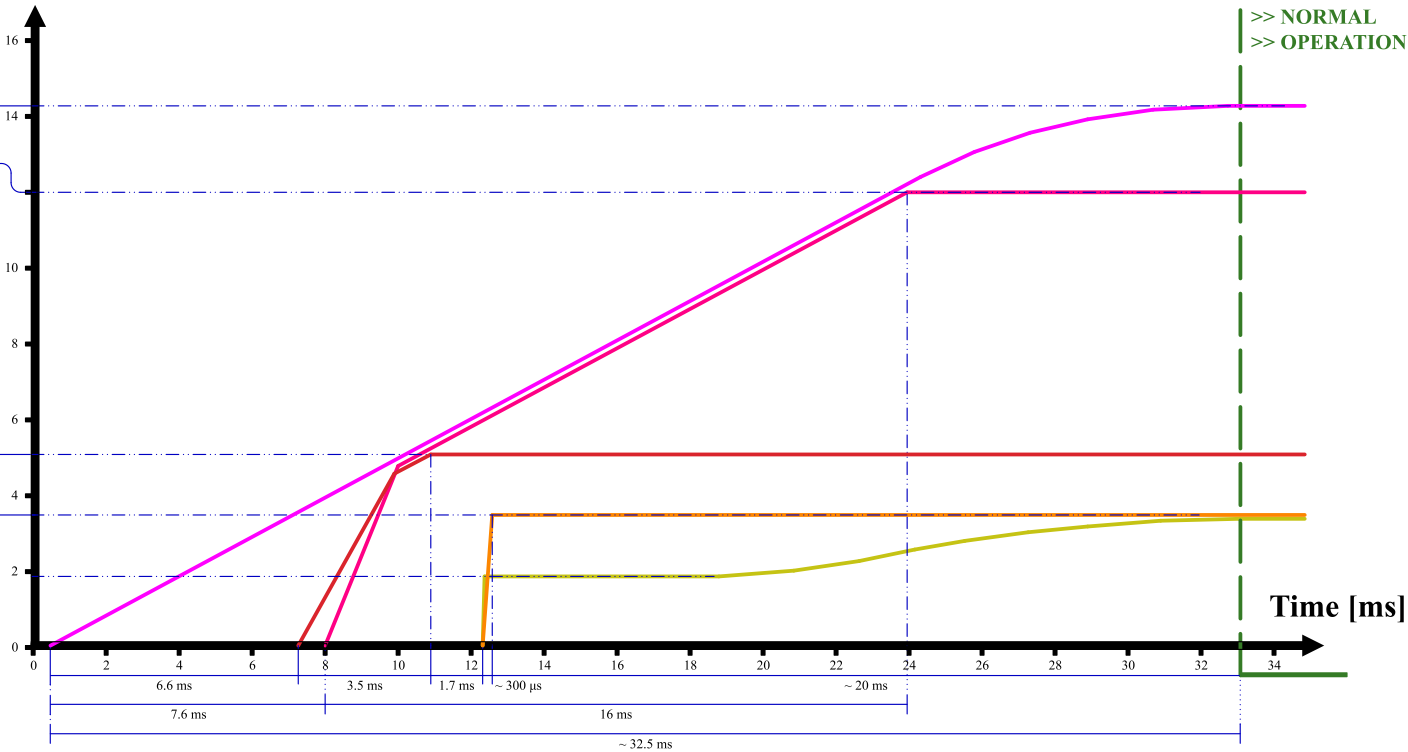
[21] Misc - Board Version, Switch



Comments:	Company: Willas Willas Electronic Corp.		Variant: DRAFT		Git Hash: e99a2fe
	Board Name: Motor Control Board			Project Name: Master Hand	
Sheet Title:	File Name: Misc - Board Version, Switch.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
Sheet Path: /Project Architecture/Misc - Board Version, Switch/		Reviewer:	Size: A4	Sheet: 21 of 23	

NAME	SOURCE	LEVEL
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%
+V??	??	? - ? ± ??%

Voltage [V]



Comments:	Company: Willas		Variant: DRAFT	Git Hash: e99a2fe
	Board Name: Motor Control Board		Project Name: Master Hand	
Sheet Title:	File Name: Power - Sequencing.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)
Sheet Path: /Power - Sequencing/		Reviewer:	Size: A4	Sheet: 22 of 23

Comments:	Company: Willas Willas Electronic Corp.		Variant: DRAFT		Git Hash: e99a2fe
	Board Name: Motor Control Board			Project Name: Master Hand	
Sheet Title:	File Name: Revision History.kicad_sch	Designer: Damon	Date: 2026-06-27	Revision: + (Unreleased)	
Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 23 of 23	